

L1

Big Picture - Road Ahead

- earlier we dived deep into T5. We have to grip them into families. Inc → dataset, scale
- 1 method → enc., dec. & enc-dec.


```

graph TD
    Enc["enc."] --> BERT_Roberta["BERT  
Roberta"]
    Dec["dec."] --> GPT3_Llamas_Gopher_PaLM["GPT3  
Llamas, Gopher,  
PaLM"]
    EncDec["enc-dec."] --> BART_T5_GLM_Switch_MT5_FlanT5["BART,  
T5, GLM,  
Switch, MT5,  
FLanT5"]
      
```
- Scaling laws → build larger models now.
- instruct GPT proposed RLHF-based alignment objectives for conversational models
- the no. of both open & closed LLMs is still growing. So how do they differ in architecture.
- the dataset used to train the model (C4 was common amongst them), various design choices to reduce computational budget, inc. the performance
- now, we prepare a unifying design.
- GPT1 → bookcorpus data
 - absolute PE, MHA, Layer norm, Gelu FFNN
 - CLM obj., adam optimizers, N=12
- GPT3 → data: CC, web, wiki, book
 - Layer Norm, sparse attenⁿ, Same PE
 - same actⁿ, obj., AdamW optimizer, N=96
- first, we will study abt. data sources, data pipelines & datasets

L2

Motivation for scaling the data size

GPT - 0.8 B words

BERT - 3.3 B words

GPT - 19 B tokens

T5 - 175 B tokens

- how to source the data? and in large amt. of data, how to check it?
- 2 inpt. $\rightarrow$ T5 & scaling laws paper
- given the no. of param-embeddings
params N & size of dataset D , test loss

$$L(N, D) = \left[\frac{N_c}{N} \alpha_N + \frac{D_c}{D} \right] \alpha_D$$

empirically
proved

$$N_c = 8.8 \times 10^{13} = 88T \text{ params}$$

$$D_c = 5.4 \times 10^{13} = 54T \text{ tokens}$$

$$\alpha_N = 0.076$$

$$\alpha_D = 0.095$$

- so actually both N & D should be increased appropriately to drive the test loss towards 0.
- emphasize diversity (Pile corpus) study is inpt. data quality is inpt. (curated web dataset by Falcon). Open source data is good enough (shown by Llama)
- scale, quality & diversity
all 3 inpt. for datasets

L8.3 Data Sourcing & Cleaning

- we have PB data → source data (100TB) → clean data - deduplicate, quality filtering (1-11TB) → release at this level. → train the LLM (1-3T tokens)
- the data cleaning pipeline is present for only few of the datasets. Datasets often lack proper documentation. For eg. MassiveWeb used in training Gopher model used a custom HTML scraper, and Google SafeSearch signal for filtering.
- always crawl the web → custom HTML scraper or use CC.
- raw data has duplicates, toxic, low quality content and also PII.
- raw CC has 3 types → WARC, WET, & WAT.
- to build a large snapshot, researchers often take multiple snapshots of the CC.
- finally, we do our best to clean the dataset.

L8.4 Pre-processing pipeline

- sug. following mechanisms
 1. detect lang.
 2. deduplicate the content
 3. detect toxicity level.
 4. to detect quality of the content

5. to detect & remove PII

- deduplicaⁿ → exact → suffix array
 → fuzzy (near)
 ↳ line-level. ↳ minHash, Simhash
- CCNet — high quality monolingual
 pre-training data
 ↳ paragraph-level deduplicaⁿ, fuzzy,
 lang. detecⁿ, target quality filtering
 → by using this, they extracted documents
 for 130 languages.
- C4 — lang. detecⁿ, 9 filters, deduplicaⁿ
 ↳ per domain-level filter, deduplicaⁿ
 ↳ erroneous gibberish, boiler plate text,
 offensive lang., low quality texts
- generally, first do fuzzy duplicaⁿ, then
 exact deduplicaⁿ.
- Refried Web → jsText, lang. identificaⁿ;
 URL filtering, quality filtering, fuzzy,
 exact
- so far all abt. diff. forms of text data.
 what abt. audio & OCR text? ⇒ SETU pipeline
 for Sangeha dataset.
- none of them is perfect
- if it is not deduplicated —
 ↳ allows the model to memorize, gives
 a wrong picture of test performance,
 susceptible to privacy risks.
- all pipelines are leaky.

LS

Pre-training datasets

- monolingual - book, wiki, codes, etc.
- multi-lingual - sangraha, etc
- compare by size of datasets as per uncompressed storage memory, or by no. of tokens
- C4 - eng, 156 B, No, web
- Pile - eng, 170 B, Yes, 22
- Stack - code, >1T, Yes, 380 PL
- RedP. - Eng/Multi, 12/30 T, Yes, quite good
- mC4 - multi, 418 B, No, webpage
- sangraha - " , 251 B, Yes, all
- composite datasets - Pile, redPajama, dolma
- data diversity - utilizing high quality & diverse datasets is crucial for achieving optimal downstream performance
- 15 mill steps (2^{16} tokens) — 1T tokens
- obs. sampling is used.